GLADYS FONG

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EDUCATION

The George Washington University, Washington, DC August 2026

Bachelor of Science: Engineering Management & Systems Engineering

Minor: Data Analytics for Decisions

Relevant Coursework: Systems Architecture, MBSE, Data Analytics, Risk Analysis, Project Management, CAD/CAM, Differential Equations, Machine Learning

SKILLS & INTERESTS

Tools & Languages: Python (Pandas, NumPy, Scikit-learn, Matplotlib), SQL, MATLAB, Capella (MBSE), SysML, SOLIDWORKS, AutoCAD, Raspberry Pi 4, GIS/Geospatial Mapping, Tableau, Excel, OpenAI API

Methods: Predictive Modeling, Data Architecture, Nonlinear Regression, Requirements Analysis, Agile/Scrum, Risk Analysis, Stakeholder Engagement, AI Agent Development

Languages: English & Spanish (Native)

AWARDS & RECOGNITIONS

- Best Paper Winner, Andrew P. Sage Memorial Capstone Competition (2026) — WMATA Safety Data Architecture & Forecasting
- IEEE Published Author for transit safety data architecture research
- Honorable Mention, SIEDS UVA Systems Engineering Competition • Best GWU SEAS Senior Design Award, Systems Engineering Department

TECHNICAL PROJECTS

WMATA Safety Data Architecture & Forecasting — Capstone Fellow Aug 2024 – May 2026

GWU x Washington Metropolitan Area Transit Authority, Washington, DC

- Redesigned WMATA's siloed safety data into a centralized logical data architecture to improve cross-departmental data accessibility and enable systematic incident analysis across rail and bus operations.
- Built a Python nonlinear regression forecasting and mapping tool that modeled seasonality and location-based incident patterns, enabling predictive maintenance planning and resource allocation.
- Led weekly stakeholder coordination meetings with WMATA safety engineers and transit officials, managing sprint planning, technical documentation, and milestone delivery across a 4-person team.

IoT Plant Health Monitoring System — Lead Engineer Jan – Apr 2026

George Washington University, Washington, DC

- Designed and deployed an embedded smart monitoring system to automate plant health tracking in controlled environments, using 4 environmental sensors, Raspberry Pi 4, and automated pump control to reduce manual monitoring effort.
- Developed ML-based plant health classification model and real-time monitoring dashboard, improving response time to environmental changes.
- Modeled full system lifecycle — operational, logical, and physical architectures — using SysML and Capella to ensure traceability and design verification throughout development.

FEMA Equitable Disaster Resource Allocation Tool — Lead Engineer Jan – May 2025

George Washington University, Washington, DC

- Built a Python tool normalizing socioeconomic, geographic, and infrastructure indicators into a Vulnerability Index to identify communities in the Houston, Texas area most at risk of inadequate disaster response based on Hurricane Harvey data.
- Produced geospatial heat maps of Houston-region neighborhoods highlighting underserved communities with limited access to emergency resources; presented findings and data-driven recommendations to FEMA Response leadership to inform future resource distribution strategies.

Electric Scooter Infrastructure Requirements Analysis Jan – Apr 2025

George Washington University, Washington, DC

- Conducted full requirements analysis for ADA-compliant e-scooter docking infrastructure in Foggy Bottom (a dense urban neighborhood in Washington, DC), addressing last-mile transit gaps for GWU students and local residents.
- Built GIS heat maps using Python, visualizing ridership demand and accessibility gaps; translated stakeholder-driven requirements into infrastructure specifications to inform urban mobility planning decisions.

Mechanical Bird Manufacturing — Systems Design & Fabrication Aug – Nov 2023

George Washington University, Washington, DC

- Designed and manufactured a mechanical bird with flapping wings and 360-degree head rotation using SOLIDWORKS, AutoCAD, and machining tools.

- Applied SysML decomposition and iterative design verification to validate functional requirements and ensure structural integrity at each stage of the fabrication process.

WORK EXPERIENCE

Undergrad Teaching Assistant — Engineering Computations (4 Terms) Jan 2024 – Dec 2026
George Washington University, Washington, DC
- Support 60+ students in computational thinking, differential equations, linear algebra, and Fourier analysis.
- Integrated AI-enhanced learning tools (MentorAI) into engineering instruction, contributing to improved student engagement and comprehension in weekly lab sessions.
- Guide students through complex topics including differential equations, linear algebra, and Fourier analysis using Python.
- Help students develop practical skills in engineering computation and mathematical modeling.

Community Support Assistant Aug 2023 – May 2026
George Washington University, Washington, DC
- Manage communications, safety inspections, and event logistics for a 200+ student residential community, serving as the primary point of contact between residents and university housing administration.
- Coordinate student programming events, enforce building access and safety protocols, and support departmental operations, including incident reporting and policy compliance.

Mathematics Tutor Aug 2020 – June 2021
West Covina, CA
- Conducted private tutoring sessions for 8+ middle and high school students, specializing in algebra, fostering a deeper understanding of mathematical concepts.
- Tailored personalized lesson plans based on each student's unique skill level, ensuring an adaptive and effective learning experience.
- Conducted weekly progress assessments to monitor individual performance, adjusting teaching methods in response to specific challenges; students demonstrated measurable grade improvements of one letter grade or more over the course of tutoring.
- Encouraged open communication, actively listening to students' concerns, and adapting instructional methods to meet their evolving needs.

LEADERSHIP & PROFESSIONAL DEVELOPMENT EXPERIENCE

Scholar April 2020 – Present
Thrive Scholars
- Selected as one of 100 high-achieving, underrepresented, first-generation students for a prestigious 6-year program focused on college access, success, and professional development.
- Actively participating in a four-year Career Development Program with personalized expert career coaching.
- Participated in a case study project at Summer Academy to build a hypothetical COVID-19 vaccine distribution plan using data analysis and systems design for Palantir, proposing an equitable allocation model that prioritized underserved communities through targeted funding and resource distribution strategies.

Scholar June – Aug 2021
Yale Young Global Scholars, Yale University
- Participated in an intensive intellectual enrichment program, developing collaborative relationships with student researchers from over 40 countries.
- Collaborated with a diverse team of scholars to research and analyze the global issue of climate-driven food insecurity, synthesizing data on supply chain disruptions and agricultural vulnerabilities across developing regions.
- Presented policy solutions to professors and YYGS faculty leaders, proposing adaptive agricultural investment strategies and regional food reserve frameworks as actionable responses to the identified crisis.

Coding Assistant July 2021 – August 2021
Kids Teach Tech, UC Berkeley
- Facilitated engaging virtual learning experiences for 1st-8th-grade students using the Scratch coding application.
- Provided extensive assistance by attending to student inquiries and providing clear tutorials on coding principles.
- Assigned and directed students through hands-on assignments that allowed them to apply newly acquired coding abilities efficiently.
- Created a collaborative and interactive learning environment to ensure students obtained practical knowledge and confidence in coding.